3 | Agents, memory and MCP

The loop

create_agent combines a model, tools and repeated decisions. Bind the authenticated identity in host context. Tools must check ownership independently of model obedience.

State and memory

A checkpointer saves a conversation under thread_id. A Store holds cross-thread data in explicit namespaces. Authorize the owner before loading either. Summaries save tokens but can omit facts.

Middleware

Hooks surround agent, model and tool execution. Use limits for bounds, retries for transient errors and fallback for defined provider failures. Count retries and nested model calls in budgets.

Human review

An interrupt pauses an action; Command(resume=...) supplies a decision. Review the exact quote and inputs. Revalidate trusted facts after resume. Rejection means no write; replay needs an idempotency key.

MCP trust

MCP transports tool schemas and results; it does not make a server trustworthy. Review program/endpoint, tool names, fields and permissions. The pinned current adapter is beta. Keep approvals outside model tools.

Failure to check

Cross-owner access, guessed approval receipts, changed action payloads, duplicate writes, runaway tools and raw sensitive tracing all need host-level tests.

Try it

```
uv sync --extra agents
uv run --no-sync python capstones/05_personal_ops/solution.py --offline
uv run --no-sync pytest tests/test_security.py
```